



PAK-MED STUDY CORNER

Your Success, Our Mission



TOTAL MCQS
180

GRAND MOCK TEST – 180 MCQs

BIOLOGY | CHEMISTRY | PHYSICS | ENGLISH

TEST
2

- 1** She would enjoy _____ among them.
A. live B. lived
C. to live D. living
- 2** After she _____ the stairs, her heart almost gave out from exhaustion.
A. has climbed B. have climbed
C. had climbed D. was climbing
- 3** Every boy and every girl _____ given a prize.
A. were B. is
C. are D. have been
- 4** The committee _____ made its final decision.
A. have B. is to
C. has D. are to
- 5** Choose the correct option.
A. They study at home every day diligently.
B. They study diligently every day at home.
C. They study at home diligently every day.
D. They study diligently at home every day.
- 6** Choose the correct option.
A. She likes reading books, swimming, and to paint.
B. She likes reading books, swimming, and painting.
C. She likes to read books, swimming, and painting.
D. She likes read books, swim, and paint.
- 7** Choose the correct option.
A. The rooms of this hotel are cleaner than a guest house.
B. The rooms of this hotel are cleaner than those of a guest house.
C. The rooms of this hotel are cleaner than guest house's.
D. The rooms of this hotel are cleaner than these of a guest house.

- 8** Despite the complicated situation, she remained _____, calmly assessing her options before deciding.
A. composed B. erratic
C. frantic D. hysterical
- 9** I _____ him only one letter up to now.
A. sent B. have sent
C. send D. will send



180 HIGH-YIELD MCQS

Carefully selected from PMDC syllabus



COVERS 4 MAJOR SUBJECTS

Biology, Chemistry, Physics & English



REAL EXAM EXPERIENCE

Timed Test | Instant Result | Smart Analysis



IMPROVE & SCORE HIGH

Practice more, perform better!

**"DISCIPLINE TODAY,
DOCTOR TOMORROW!"**



**STUDY SMART
SCORE HIGH**



**FREE NOTES
PAST PAPERS**



**DAILY TESTS
MOCK TESTS**



**ACHIEVE YOUR
DREAM MBBS**

16 The number of nuclear pores in differentiated cells such as erythrocytes have only:

- A) 10 to 20 pores/nucleus
- B) 5 or 10 pores/nucleus
- C) 1 to 2 pores/nucleus
- D) 3 or 4 pores/nucleus

17 Following processes can be carried out without involving membrane proteins, EXCEPT:

- A) Receptor-mediated endocytosis
- B) Pinocytosis
- C) Exocytosis
- D) Phagocytosis

18 Secretory granules (transport vesicles) are pinched off at the _____ face of Golgi apparatus:

- A) Forming face
- B) Maturing face
- C) Cis face
- D) Tran face

19 Unlike animals both prokaryotes and plants have:

- A) Cell membrane
- B) Cell wall
- C) Ribosomes
- D) Plasmids

20 The finishing, packing and distribution section of the cell is:

- A) ER
- B) Lysosome
- C) Peroxisome
- D) Golgi apparatus



Golgi Apparatus

21 The fluid located in the center of an animal cell is called:

- A) Cytoplasm
- B) Nucleoplasm
- C) Cell sap
- D) Cytosol

22 The presence of ribosomes and DNA indicates that mitochondria also synthesize some:

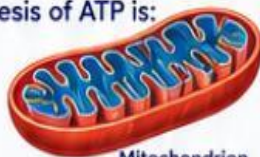
- A) Carbohydrates
- B) Lipids
- C) Proteins
- D) ATP

23 Previously the organisms were divided into:

- A) Prokaryotes and eukaryotes
- B) Animals and plants
- C) Bacteria and fungi
- D) Autotrophs and heterotrophs

24 Organelle involved in the synthesis of ATP is:

- A) Ribosome
- B) Mitochondrion
- C) Nucleus
- D) Centriole



Mitochondrion

25 A barrier between the cell contents and its environment is:

- A) Cell wall
- B) Cell membrane
- C) Nuclear membrane
- D) Tonoplast

26 _____ controls the life and activities of the cell:

- A) Nucleus
- B) Glyoxasome
- C) Ribosome
- D) Lysosome

27 Plant cell being an autotroph lacks:

- A) Photosynthesis
- B) Phagocytosis
- C) Centriole
- D) Cytokinesis

28 Disc-shaped protein structures within the centromere to which the spindle fibres are attach during mitosis or meiosis are:

- A) Kinetochores
- B) Centromeres
- C) Centrosomes
- D) Nucleosomes

29 Structure and chemical composition of prokaryotic _____ is different from that of eukaryotes:

- A) Plasma membrane
- B) Cell wall
- C) Ribosome
- D) Nuclear membrane

30 It is a complex system of interconnected tubules around the central stacks:

- A) Golgi apparatus
- B) Golgi complex
- C) Endoplasmic reticulum
- D) Centrioles

31 Both prokaryotic and eukaryotic cell walls do not differ with respect to:

- A) Composition
- B) Peptidoglycan
- C) Structure
- D) Location

32 The material of cell is recycled and renewed by an organelle called:

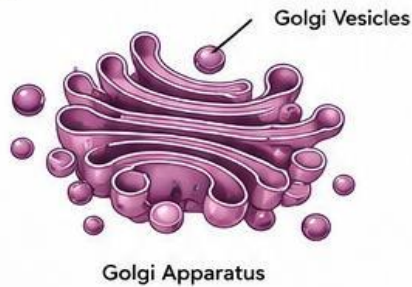
- A) Autophagosome
- B) Heterophagosome
- C) Primary lysosome
- D) Secondary lysosome

33 The carrier mediated, energy dependent movement of material into or out of cells, especially against concentration gradient is called:

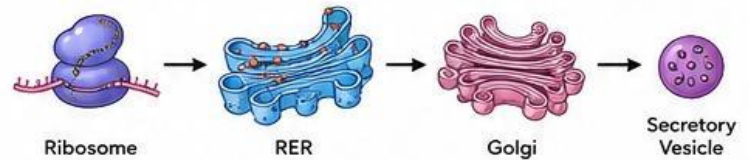
- A) Passive transport
- B) Diffusion
- C) Active transport
- D) Osmosis

- 21** Which of the following is NOT a product of Golgi complex?

- A. Chitin
- B. Cellulose
- C. Lysozyme
- D. ATP



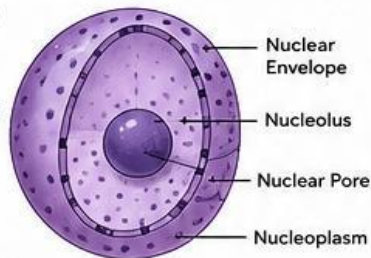
- 22** In which order does the pathway of a newly synthesized secretory protein occur?



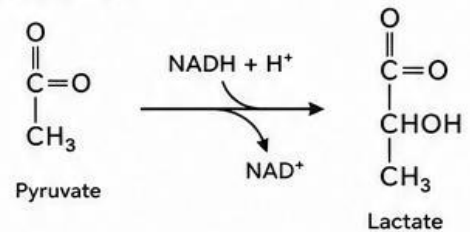
- A. Ribosome → RER → Golgi → Secretory vesicle
- B. Ribosome → SER → Golgi → Nucleus
- C. Polysome → Golgi → Lysosome → Cytoplasm
- D. Nucleus → SER → Ribosome → Plasma membrane

- 23** Which statement about nucleoplasm is TRUE?

- A. It is same in composition as cytoplasm
- B. It contains nucleotides, polymerases, etc.
- C. It lacks histone proteins
- D. It cannot pass materials through nuclear pores



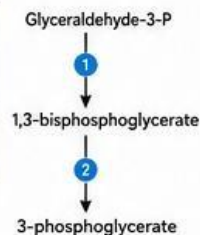
- 24** In lactic acid fermentation, pyruvate is converted into:



- A. Ethanol
- B. Acetyl-CoA
- C. Lactate
- D. $\text{CO}_2 + \text{H}_2\text{O}$

- 25** Which option fills correctly?

Step	Product	ATP/NADH ₂
1	Glyceraldehyde-3-P → 1,3-bisphosphoglycerate	?
2	1,3-bisphosphoglycerate → 3-phosphoglycerate	?



- A. 2 ATP, 1 NADH₂
- B. 1 ATP, 2 NADH₂
- C. 1 NADH₂, 1 ATP
- D. 2 NADH₂, 2 ATP

- 26** Which phase of glycolysis produces ATP?

- A. Preparatory
- B. Oxidative
- C. Both A and B
- D. None

- 27** During starvation, proteins contribute energy mainly through conversion into:

- A. Glycerol
- B. Acetyl-CoA
- C. Glucose
- D. Nucleotides

- 28** Parvoviruses are classified as:

- A. ssRNA viruses
- B. dsDNA viruses
- C. ssDNA viruses
- D. dsRNA viruses

- 29** The main target of HIV are:

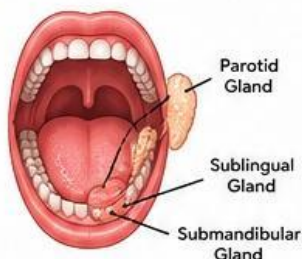
- A. RBCs
- B. Helper T-lymphocytes
- C. Neutrophils
- D. B-cells only

- 30** Which is true about AIDS stage?

- A. Appears immediately after infection
- B. May remain asymptomatic for years
- C. Never associated with cancer
- D. Can be fully cured with ART

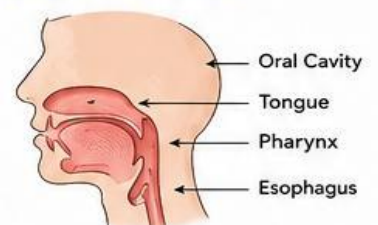
- 31** Saliva performs all EXCEPT:

- A. Lubricates food with mucin
- B. Contains amylase to break starch → maltose
- C. Contains lysozyme to kill bacteria
- D. Contains lipase to digest fats



- 32** The process of swallowing is also termed:

- A. Deglutition
- B. Peristalsis
- C. Assimilation
- D. Mastication

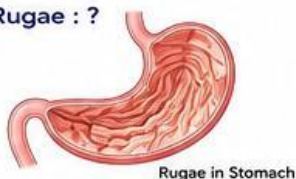


33 Which row is mismatched?

Option	Cell Type	Secretion
A.	Parietal cell	HCl + intrinsic factor
B.	Chief cell	Pepsinogen
C.	G-cells	Gastrin
D.	Mucous cells	Lipase

34 Plicae circulares : Ileum :: Rugae : ?

- A. Stomach
- B. Jejunum
- C. Duodenum
- D. Colon

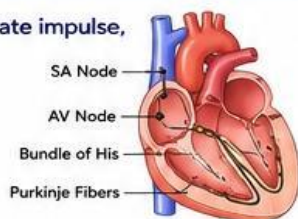


35 Arteries are best defined as the vessels which:

- A. Supply oxygenated blood to the different organs
- B. Carry blood away from the heart to different organs
- C. Break up into capillaries which reunite to form a vein.
- D. Carry blood from one visceral organ to another visceral organ

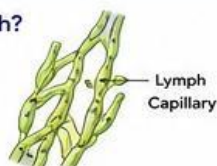
36 If SA node fails to generate impulse, the next pacemaker is:

- A. Bundle of His
- B. AV node
- C. Purkinje fibers
- D. Left atrium wall



37 Which one is INCORRECT about lymph?

- A. Colorless fluid with lymphocytes
- B. Transports proteins and fats
- C. Contains more RBCs than plasma
- D. Formed from interstitial fluid



38 Blood flows in veins toward heart despite low pressure due to:

- A. SA node stimulation
- B. Valves + Skeletal muscle contraction
- C. Elastic tunica media
- D. Arterial recoil

39 Which one is INCORRECT about veins?

- A. Thin walls compared to arteries
- B. Carry always deoxygenated blood
- C. Possess valves
- D. Pressure is low inside

40 Which antibody secreting cells are short-lived?

- A. Plasma cells
- B. Memory B cells
- C. TH cells
- D. TC cells

41 A patient receives antibodies for snake venom through injection. This is:

- A. Natural active immunity
- B. Artificial passive immunity
- C. Artificial active immunity
- D. Natural passive immunity

42 Which immune cells suppress the activity of Tc cells after clearance of pathogen?

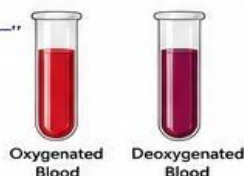
- A. Plasma cells
- B. Memory cells
- C. Ts cells
- D. NK cells

43 Identify the correct statements:

- I. MHC-I is found on all nucleated cells.
 - II. MHC-II is found only on APCs.
 - III. MHC molecules act as self-identity markers.
- A. I & II only B. II & III only C. I & III only D. I, II & III

44 Color of oxygenated blood is _____, deoxygenated blood is _____.

- A. Bright red, dark maroon
- B. Dark red, light pink
- C. Blue, red
- D. Scarlet, brown



45 Which are true for carbon dioxide transport?

- I. Mostly transported as bicarbonate
- II. 20% as carbamino hemoglobin
- III. 10% dissolved in plasma
- IV. 100% as carbonic acid

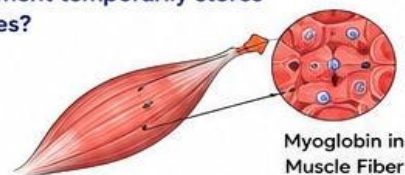
- A. I, II & III only
- B. II & IV only
- C. I & III only
- D. I, II, III & IV

46 At high altitude, breathing increases because:

- A. Air contains more CO₂
- B. Air has low O₂ partial pressure
- C. Lungs lose surfactant
- D. Alveoli fill with residual volume

47 Which blood pigment temporarily stores oxygen in muscles?

- A. Hemoglobin
- B. Myoglobin
- C. Fibrinogen
- D. Albumin



48 Which factor does NOT affect breathing rate?

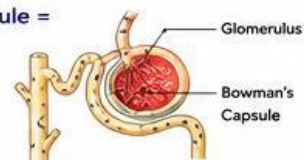
- A. Blood CO₂
- B. Blood O₂
- C. Blood O₂
- D. Blood H⁺

49 Vitamin produced by kidney that helps in calcium absorption:

- A. Vitamin A
- B. Vitamin B₁₂
- C. Vitamin D₃
- D. Vitamin E

50 Glomerulus + Bowman's capsule =

- A. Malpighian body
- B. Nephron
- C. Collecting duct
- D. Loop of Henle



51 Which ions are exchanged in counter-current system?

- A. Na⁺, Cl⁻, K⁺
- B. HCO₃⁻, H⁺
- C. Ca²⁺, Mg²⁺
- D. Urea only

52 The proximal convoluted tubule (PCT) is mainly responsible for:

- A. Filtration
- B. Reabsorption
- C. Urine storage
- D. Hormone secretion

53 Which is NOT a function of kidneys?

- A. Blood filtration
- B. Excretion of nitrogenous wastes
- C. Synthesis of Vitamin A
- D. Regulation of pH

54 The osmotic gradient in medulla is mainly by:

- A. PCT
- B. Loop of Henle
- C. Collecting duct
- D. Vasa recta

55 Osteocytes are found in small spaces called:

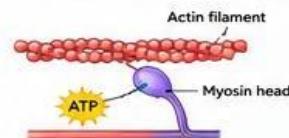
- A. Lacunae
- B. Trabeculae
- C. Osteons
- D. Canaliculi

56 Functionally cardiac muscles are similar to:

- A. Unstriated muscles
- B. Striped muscles
- C. Striped & Unstriated muscles
- D. None

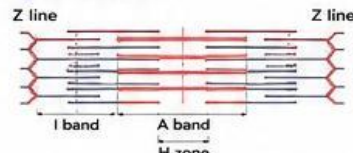
57 Cross bridge is broken due to:

- A. Binding of ATP at myosin head
- B. Hydrolysis of ATP at myosin head
- C. Binding of ATP at troponin
- D. Hydrolysis of ATP at troponin



58 Which one is not possible during contraction:

- A. I band remains same
- B. A band remains same
- C. Sarcomere length decrease
- D. H-zone disappears



59 The number of bones in the lower jaw of human is :

- A. 1 B. 4 C. 6 D. 8

60 The skull bone that articulates with the first cervical vertebra is the:

- A. Occipital bone B. Parietal bone
C. Temporal bone D. Sphenoid bone

61 Cerebellum is NOT responsible for:

- A. Balance B. Coordination
C. Intelligence D. Posture

62 Medulla controls all except:

- A. Swallowing B. Breathing
C. Heart rate D. Reasoning

63 Action potential may reach up to:

- A. +30 mV B. +50 mV
C. -90 mV D. 0 mV

64 Reflex are typically involves:

- A. 1 sensory + 1 motor neuron only
B. 1 sensory + 1 interneuron + 1 motor neuron
C. CNS without neurons
D. Effectors only

65 Functions of neuroglia include:

- I. Guiding axon migration
II. Immune functions
III. Supplying nutrients
A. I only B. I & II only
C. II & III only D. I, II & III

66 The myelin sheath in CNS is produced by:

- A. Schwann cells B. Oligodendrocytes
C. Neurolemma D. Astrocytes

67 Which receptor type in the hypothalamus detects changes in blood osmolarity?

- A. Olfactory receptors B. Osmoreceptors
C. Gustatory receptors D. Photoreceptors

68 Imaging technique named with mAbs

- A. MRI B. PET
C. CT D. Ultrasound

69 Which group of conditions is explicitly listed under stem cell therapies?

- A. Cataract, hepatitis, influenza
B. Parkinson's disease, spinal cord injuries, diabetes
C. TB, syphilis, Lyme disease
D. Cystic fibrosis, muscular dystrophy, sickle-cell anemia

70 Which vaccine is mismatched?

- A. Hepatitis B — made via recombinant DNA in yeast
B. HPV — gene inserted into insect cells to make protein
C. COVID-19 — uses mRNA to instruct cells to make spike protein
D. Malaria — updated yearly after WHO strain selection

71 Clinical samples compatible with probes:

- A. Only cerebrospinal fluid
B. Blood, urine, or tissue
C. Exhaled air exclusively
D. Hair and nails only

72 Analogous organs arise due to:

- A. Same structure, same ancestry
B. Different structure, same ancestry
C. Same function, different ancestry
D. Same origin always

73 Example of homologous organs:

- A. Wings of bat & insect
B. Flippers of dolphin & arm of man
C. Insect leg & cat paw
D. Wings of bird & butterfly

74 Which is a reptilian character of Archaeopteryx?

- A. Beak B. Feathers
C. Wing claws & teeth D. Keel-less sternum

75 Ancient single land mass was called:

- A. Gondwana B. Laurasia
C. Pangaea D. Panthalassa

76 The acrosome of sperm is derived from:

- A. Lysosome B. Golgi body
C. Nucleus D. Centriole

77 Rete testis connects:

- A. Seminiferous tubules to vas deferens
B. Seminiferous tubules to epididymis
C. Seminiferous tubules to efferent ducts
D. Seminiferous tubules to urethra

78 Which structure carries sperm into urethra through prostate gland?

- A. Vas deferens B. Ejaculatory duct
C. Epididymis D. Rete testis

79 Which of the following occurs in luteal phase?

- I. Formation of corpus luteum
II. Secretion of progesterone
III. Preparation of endometrium
IV. Shedding of endometrium
A. I, II, III only B. II & IV only
C. I & III only D. All four

80 Ovulation typically occurs on day:

- A. 7 B. 14 C. 21 D. 28

81 Which hormone rises during follicular phase?

- A. FSH B. Progesterone
C. LH D. Inhibin

82 The phase dominated by estrogen is:

- A. Secretory B. Proliferative
C. Menstrual D. Ovulatory

83 Seed shape trait (round vs wrinkled) showed F_2 ratio approximately:

- A. 2.5:1 B. 2.96:1
C. 4:1 D. 1:1

84 Yellow cotyledon $\times$ green cotyledon in F_2 showed ratio:

- A. 9:3:3:1 B. 3:1
C. 2:8:1 D. 1:2:1

85 If R = round and r = wrinkled, F_2 generation of $Rr \times Rr$ gives wrinkled seeds with probability:

- A. 1/4 B. 2/4
C. 3/4 D. 0

BIOLOGY

Tall (TT) × dwarf (tt) produces all:

- A. Tall heterozygous
- B. Dwarf homozygous
- C. Mixed tall and dwarf
- D. Tall homozygous

Incomplete dominance produces:

- A. New blended phenotype
- B. No recessive character
- C. Equal dominance
- D. Independent assortment

If father has genotype I^Ai and mother has I^Bi, child may have blood group:

- A. A, B only
- B. A, B, AB, O
- C. AB only
- D. O only

The daughters of a hemophilic man and a normal woman will be:

- A. All normal
- B. All carriers
- C. All hemophilic
- D. 50% carriers

If the alleles for seed shape (R/r) and seed color (Y/y) are on different chromosomes, the expected phenotypic ratio in the F_2 generation is:

- A. 9:3:3:1
- B. 1:2:1
- C. 3:1
- D. 1:1:1:1

CHEMISTRY

The IUPAC name for isobutyl chloride is:

- A. 2-Methyl-2-chloro butane
- B. 2-Chloro-2-methyl butane
- C. 1-Chloro-2-methyl propane
- D. 2-Methyl-3-chloro propane

How many structural isomers are possible for C₅H₈ having one triple bond?

- A. 4
- B. 3
- C. 5
- D. 1

The main difference between dyes and pigments is:

- A. Dyes dissolve in water; pigments are insoluble
- B. Pigments are always natural
- C. Dyes are used only in plastics
- D. Pigments cannot provide color

Dyes are substances used to:

- A. Strengthen materials
- B. Impart permanent color
- C. Increase conductivity
- D. Remove stains

Denatured proteins lose their:

- A. Biological activity
- B. Peptide bonds
- C. Amino acids
- D. Carbon content

Hydrolysis of esters produces:

- A. Acid + Alcohol
- B. Ketone + Alcohol
- C. Amine + Alcohol
- D. Aldehyde + Alcohol

Hydrolysis of acyl halides gives:

- A. Acid anhydrides
- B. Carboxylic acids
- C. Alcohols
- D. Ketones

The product of reducing formaldehyde with H₂/Ni is

- A. Methanol
- B. Method
- C. Ethanol
- D. Formic acid

Aldehydes are more reactive than ketones because:

- A. Ketones have no α -hydrogen
- B. Aldehydes have less steric hindrance & less +I effect
- C. Aldehydes are less polar
- D. Ketones are always aromatic

In which of the following reactions of alcohol there is no cleavage of C–O bond?

- A. Oxidation reaction of alcohol
- B. Dehydration reaction of alcohol
- C. Reduction reaction of alcohol
- D. Reaction of alcohol with phosphorous tribromide

When phenol is treated with dilute bromine water, the major product is:

- A. o-bromophenol
- B. p-bromophenol
- C. 2,4,6-tribromophenol
- D. m-bromophenol

102

Ketones are first oxidation products of:

- A. Primary alcohols
- B. Secondary alcohols
- C. Both
- D. None

103

During elimination, the order of reactivity of halides is:

- A. 3° > 2° > 1°
- B. 1° > 2° > 3°
- C. 2° > 3° > 1°
- D. 3° > 1° > 2°

104

The given reaction belongs to which type:



- A. S_N1 reaction
- B. S_N2 reaction
- C. E1 reaction
- D. E2 reaction

105

Which of the following statements is incorrect?

- A. Alkanes undergo substitution reactions under UV light.
- B. Alkanes dehydrogenate on heating with Ni at 500°C to give alkenes.
- C. Alkanes can be aromatized by heating with Pt/Cr₂O₃.
- D. Alkanes generally undergo electrophilic addition reactions.

A. A and B only

B. A, B and C only

C. D only

D. All A, B, C, D

106

The order of reactivity of alkenes towards electrophilic addition is:

- A. Ethene > Propene > 2-methylpropene
- B. 2-methylpropene > Propene > Ethene
- C. Propene > Ethene > 2-methylpropene
- D. Ethene > 2-methylpropene > Propene

107

Which reagent will convert benzene into phenol in one step?

- A. KMnO₄
- B. NaOH (aq) at room temp.
- C. NaOH (aq), 300°C, 300 atm
- D. H₂O₂

108

Which quantum number explains the size of an orbit?

- A. Principal quantum number (n)
- B. Azimuthal quantum number (l)
- C. Magnetic quantum number (m)
- D. Spin quantum number (s)

109

Electronic configuration of oxygen (Z=8) is:

- A. 1s²2s²2p⁴
- B. 1s²2s²2p⁶
- C. 1s²2s²2p²

110

In an electrochemical cell, the cathode is defined as the electrode:

- A. Where oxidation always occurs
- B. Which always carries negative charge
- C. Where reduction takes place
- D. Whose potential is always positive

111

Why is the Standard Hydrogen Electrode chosen as a reference electrode?

- A. It can act only as a cathode
- B. Its potential is universally measurable
- C. Its potential remains constant under standard conditions
- D. It is the easiest electrode to construct

112

The correct options for the rate law that corresponds to an overall first order reaction is:

- A. Rate = k²
- B. Rate = k
- C. Rate = k^{1/2}
- D. Rate = k^{1/2} 1/2

- 113** The rate equation for the reaction $2A + B \rightarrow C$ is found to be: rate = k . The correct statement in relation to this reaction is that the
- Rate of formation of C is twice the rate of disappearance of A.
 - Half life is a constant
 - Unit of k must be s^{-1}
 - Value of k is independent of the initial concentrations of A and B

- 114** For combustion of 44 g of CO_2 , oxygen required is:
- 22.4 L
 - 44.8 L
 - 11.2 L
 - Zero

- 115** The maximum amount of CO_2 obtained by complete combustion of 2 mol C and 1 mol O_2 is:
- 1 mol
 - 2 mol
 - 3 mol
 - 0.5 mol

- 116** When a system is at Equilibrium:
- The mass of the product is equal to the mass of the reactant
 - The ratio of the product of the mass of the products and that of the reactants is constant
 - The ratio of the velocity of the forward reaction and the backward reaction is one
 - Number of moles of the product and that of the reactant are the same

- 117** For which of the following gaseous equilibrium at constant temperature, doubling the volume would cause a shift of equilibrium to the right?
- $2CO + O_2 \rightleftharpoons 2CO_2$
 - $N_2 + 3H_2 \rightleftharpoons 2NH_3$
 - $N_2O_4 \rightleftharpoons 2NO_2$
 - $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

- 118** The oxidation state of +1 for phosphorus is found in:
- Phosphorous acid (H_3PO_3)
 - Orthophosphoric acid (H_3PO_4)
 - Hypophosphorous acid (H_3PO_2)
 - Hypophosphoric acid ($H_4P_2O_6$)

- 119** A reducing agent is a substance :
- In which an element undergoes increase in oxidation number.
 - In which an element undergoes decrease in oxidation number.
 - which gains electron(s)
 - which shares electron(s)

- 120** For the process to occur under adiabatic conditions, the correct condition is:
- $\Delta T = 0$
 - $\Delta p = 0$
 - $q = 0$
 - $w = 0$

- 121** A thermodynamic state function is a quantity :
- Used to determine the heat changes
 - Whose value is independent of path
 - Used to determine work
 - Whose value depends on path

- 122** Number of σ and π bonds present in $CH_2 = CH - CH = C \equiv CH$ are:
- 10 σ , 3 π
 - 10 σ , 2 π
 - 9 σ , 2 π

- 123** By hybridization, we mean the mixing of:
- Electrons
 - Atomic orbitals
 - Atoms
 - Protons

- 124** If the pressure of a gas is doubled at constant temperature, its volume becomes:
- Doubled
 - Halved
 - Constant
 - Zero

- 125** The temperature $-273^\circ C$ is called:
- Freezing point of water
 - Boiling point of liquid nitrogen
 - Absolute zero
 - Standard temperature

- 126** Vapor pressure of a liquid increases with:
- Decrease in temperature
 - Increase in temperature
 - Increase in surface area
 - Addition of solute

- 127** Boiling point of a liquid is the temperature at which:
- Its vapor pressure equals atmospheric pressure
 - It evaporates rapidly
 - Surface tension becomes zero
 - Diffusion stops

- 128** Diamond is an example of:
- Covalent network solid
 - Molecular solid
 - Metallic solid
 - Ionic solid

- 129** The three-dimensional arrangement of particles in a crystal is called:
- Unit cell
 - Crystal lattice
 - Motif
 - Structure factor

- 130** The compound that is most reactive towards electrophilic nitration is:
- Toluene
 - Benzene
 - Benzoic acid
 - Nitro benzene

- 131** Phosphorus oxides depend on oxygen supply. Limited O_2 gives P_2O_3 ; excess O_2 gives:
- P_2O
 - P_2O_3
 - P_4O_2
 - P_4O_{10}

- 132** Among group-1A metals, the oxide category trend down the group ($Li \rightarrow Cs$) is best summarised as:
- superoxide $\rightarrow$ peroxide $\rightarrow$ oxide
 - oxide $\rightarrow$ peroxide $\rightarrow$ superoxide
 - peroxide $\rightarrow$ oxide $\rightarrow$ superoxide
 - oxide $\rightarrow$ superoxide $\rightarrow$ peroxide

- 133** The formation of the oxide ion O^{2-} requires first an exothermic and then an endothermic step as shown below :
- $$O(g) + e^- = O^-(g) \quad \Delta H^\circ = -142 \text{ kJ mol}^{-1}$$
- $$O^-(g) + e^- = O^{2-}(g) \quad \Delta H^\circ = +844 \text{ kJ mol}^{-1}$$
- This is because :
- O^- ion will tend to resist the addition of another electron
 - Oxygen has high electron affinity
 - Oxygen is more electronegative
 - O^- ion has comparatively larger size than oxygen atom

- 134** Across a period, atomic radius decreases mainly due to
- Higher effective nuclear charge
 - Addition of shells
 - Decrease in ionization potential
 - Increase in shielding

- 135** Which element has nearly same radius as Sn due to lanthanide contraction?
- Ga
 - Pb
 - In
 - Tl

136 The scalar product is zero when the angle between vectors is:

- A. 0° B. 90° C. 180° D. 45°

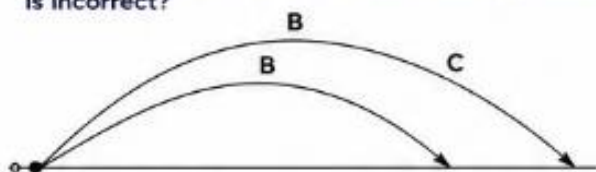
137 Area of a parallelogram formed by two vectors is given by:

- A. $|A \cdot B|$ B. $|A \times B|$ C. $A + B$ D. None

138 A bird flies due North with velocity 20 ms^{-1} for 15 s it rests for 5 s and then flies due South with velocity 24 ms^{-1} for 10 s. Find the average speed and magnitude of average velocity. For the whole trip.

- A. 18 m/s, 3 m/s B. 12 m/s, 4 m/s
C. 9 m/s, 2 m/s D. 18 m/s, 2 m/s

139 Three projectiles A, B and C are thrown from the same point in the same plane. Their trajectories are shown in the figure. Which of the following statement is incorrect?



- A. The launch speed is same for all the three
B. The time of flight is the same for all the three
C. The horizontal velocity component is largest for particle C
D. The maximum height is same for all the three

140 The energy of a photon is doubled. Its wavelength becomes:

- A. Doubled B. Halved C. Same D. Zero

141 The work-function of a photo-electric material is 3.3 eV. The threshold frequency will be equal to

- A. $8 \times 10^{14} \text{ Hz}$ B. $5 \times 10^{14} \text{ Hz}$
C. $8 \times 10^{12} \text{ Hz}$ D. $4 \times 10^{12} \text{ Hz}$

142 The energy of an orbit is proportional to:

- A. $+1/n$ B. $-1/n$ C. $-1/n^2$ D. $+n^2$

143 The uniqueness of spectral lines of each element is comparable to:

- A. Boiling point B. Fingerprint
C. Density D. Colour of flame

144 Which property remains unchanged in all nuclear decays?

- A. Mass number
B. Total energy of the system
C. Number of nucleons in each product
D. Individual atomic number of each particle

145 If a sample loses 75% of its nuclei in 20 minutes, its half-life is

- A. 5 min B. 10 min C. 20 min D. 40 min

146 P-type semiconductor is formed when

- (i) As impurity is mixed in Si
(ii) Al impurity is mixed in Si
(iii) B impurity is mixed in Ge
(iv) P impurity is mixed in Ge

- A. (i) and (iii) B. (i) and (iv)
C. (ii) and (iii) D. (ii) and (iv)

147 The forbidden energy band gap in conductors, semiconductors and insulators are E_{G1} , E_{G2} and E_{G3} respectively. The relation among them is -

- A. $E_{G1} = E_{G2} = E_{G3}$ B. $E_{G1} < E_{G2} < E_{G3}$
C. $E_{G1} > E_{G2} > E_{G3}$ D. $E_{G1} < E_{G2} = E_{G3}$

148 Maximum and minimum force on the wire occur respectively at:

- A. 0° , 90° B. 90° , 0°
C. 45° , 0° D. 90° , 45°

149 If a positive charge enters parallel to the magnetic field then which of the following are true.

- I. Particle will move in straight line path
II. No force will act on the particle
III. The particle may reverse it's direction of motion
IV. All of the above

- A. I Only B. I, II Only
C. I, II, III D. I, II, III, IV

150 In which situation will no emf be induced in a coil?

- A. Coil is rotated in a uniform magnetic field
B. Coil is moved parallel to the direction of magnetic field
C. Magnetic field through coil changes with time
D. Coil's area is changed in a magnetic field

151 A transformer transforms 220 volts to 11 volts. If the current strength in the primary coil is 5 amp, and that in the secondary, 90 amp, what is the efficiency of the transformer

- A. 90%
B. 100%
C. 211%
D. 150%

152 Using Ohm's law with $v = V_0 \sin(\omega t)$ across a resistor R, the current is

- A. $i = (V_0 / R) \sin(\omega t)$
B. $i = (V_0 / R) \cos(\omega t)$
C. $i = V_{\text{rms}} / R$ (constant)
D. $i = 0$

153 One cycle of a sine-wave AC quantity is defined as:

- A. Any interval of 1 second
B. One positive half only
C. One complete set of positive and negative values
D. The time for current to reach its peak

154 In a thermodynamic process pressure of a fixed mass of a gas is changed in such a manner that the gas releases 20 joules of heat and 8 joules of work was done on the gas. If the initial internal energy of the gas was 30 joules, then the final internal energy will be:

First Law of Thermodynamics:

$$\Delta U = q + w$$

where,

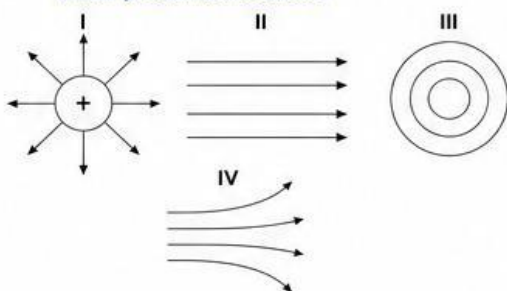
- q = heat added to the system (+)
or rejected from the system (-)
 w = work done on the system (+)
or by the system (-)

- A. 2 J B. 42 J C. 18 J D. 58 J

155 If an ideal gas is compressed during isothermal process then :

- A. No work is done against gas
- B. Heat is rejected by gas
- C. It's internal energy will increase
- D. Pressure does not change

156 The figures shows electrostatic lines of force, which pattern is incorrect :



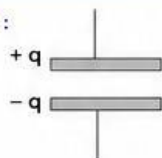
- A. I, III
- B. II, III
- C. Only IV
- D. III, IV

157 The ratio of electrostatic force to the gravitational force between them is of the order of:

- A. 1.23×10^{37}
- B. 1.23×10^{38}
- C. 1.23×10^{40}
- D. 1.23×10^{42}

158 The net charge on a capacitor is :

- A. $2q$
- B. $q/2$
- C. 0
- D. infinity



159 A capacitor is a perfect insulator for:

- A. Direct current
- B. Alternating current
- C. Both for the direct and alternating current
- D. Rapidly fluctuating current

160 Resistivity of a conductor depends primarily on:

- A. Length and area
- B. Material and temperature
- C. Current through it
- D. Applied voltage

161 Which of the following have the same temperature coefficient of resistivity?

- A. iron and silver
- B. platinum and silver
- C. iron and platinum
- D. silver and gold

162 If the linear momentum of the object is increased by 0.1%, then the kinetic energy is increased by:

- A. 0.1 %
- B. 0.4 %
- C. 0.2 %
- D. 0.01 %

163 A person carries a load horizontally. Work done against gravity is:

- A. Positive
- B. Zero
- C. Negative
- D. Maximum

164 What is the circumference of a circle, having radius of 50cm?

- A. 3.12 m
- B. 3.16 m
- C. 3.14 m
- D. 3.18 m

165 When the terminal velocity is reached, the acceleration of a body moving through viscous medium is

- A. Zero
- B. Negative
- C. Positive
- D. Equal to 'g'

166 If two points on a rotating disc have different radii, which statement is true?

- A. They have the same linear displacement
- B. The point with larger radius has greater linear displacement
- C. The point with smaller radius has greater linear displacement
- D. The point with greater radius has smaller linear displacement

167 The lift of aeroplane is based on:

- A. Torricelli's theorem
- B. Bernoulli's Principle
- C. Law of gravitation
- D. Coulomb's Law

168 If the frequency of a wave increases, its time period:

- A. Increases
- B. Remains constant
- C. Decreases
- D. Becomes zero

169 At 0°C, the speed of sound in air is approximately:

- A. 330 m/s
- B. 343 m/s
- C. 332 m/s
- D. 300 m/s

170 A particle in SHM has an amplitude of 0.2 m and angular frequency 4 rad/s. What is its maximum velocity?

- A. 0.8 m/s
- B. 1.6 m/s
- C. 0.5 m/s
- D. 0.2 m/s

171 The low resistivity of copper makes it:

- A. A poor conductor
- B. A good conductor
- C. A good insulator
- D. A poor insulator

LOGICAL REASONING

172 Statements:

All actors are fool. All fool are illiterate.

Conclusion:

- I. All actors are illiterate.
- II. All illiterate are actors.
- III. All illiterate are fools.
- IV. Some illiterates would be actors.

- A. Only II follows
- B. Only III and IV follow
- C. Only I and IV follow
- D. None follows

173 Statements:

Some phones are watches.

No watch is white.

Conclusion:

- I. Some phones are not white
- II. Some watches are phones

- A. Only I
- B. Only II
- C. Both I and II
- D. Neither I nor II

174 Statement:

A large number of people gathered on the highway blocking the traffic movement to protest the killing of five locals by a speeding vehicle.

Courses of action:

- I) The police should fire tear gas shells to disperse the crowd.
- II) The police authority should calm down the sentiment of the crowd assuring action against the culprit and deploy police personnel at the spot.

- A. Only I follows
- B. Only II follows
- C. Both I and II follow
- D. Either I or II follows

175 Statements:

- I. Ali did not go to school today.
- II. Umer is brother of Ali and he did not go to school today.
- A. Statement I is the cause and statement II is its effect
- B. Statement II is the cause and statement I is its effect
- C. Both the statements I and II are independent causes
- D. Both the statements I and II are effects of some common cause

177 If 35796 is the code for 44887.
Then the number for the code 46823 is:

- A. 37732
- B. 55914
- C. 57914
- D. 59514

179 Statements:

No Carrot is Banana.
No Banana is Orange

Conclusions:

- I. No carrot is orange.
- II. Some oranges are carrots
- III. Some carrots are not Oranges
- A. Only I and III follows
- B. Only II follows
- C. All follows
- D. None follows

176 Statements:

- I. Government has opened two hospitals in the beginning of the year.
- II. People will get good medical attention.
- A. Statement I is the cause and statement II is its effect
- B. Statement II is the cause and statement I is its effect
- C. Both the statements I and II are independent causes
- D. Both the statements I and II are effects of independent causes

178 Waiter : Tip

- A. Student: Marks
- B. Worker: Bonus
- C. Employee: Wages
- D. Clerk: Bribe

180 Find the missing term

3, 10, 101, ?

- A. 10101
- B. 10201
- C. 10202
- D. 11012



**STAY FOCUSED, STAY CONSISTENT,
SUCCESS IS INEVITABLE!**



- ★ Hard work today, bright future tomorrow.
- ★ Don't stop until you're proud.
- ★ Every expert was once a beginner.
- ★ Believe in yourself and all that you are.

You've got this! Keep pushing forward!



JOIN OUR WHATSAPP GROUP

Get Daily MCQs, Notes, Past Papers, Updates & Much More!



CLICK THE LINK BELOW TO JOIN NOW



ANSWER KEY

Q	A	Q	A	Q	A	Q	A	Q	A
1	D	41	B	81	A	121	B	161	C
2	C	42	C	82	B	122	A	162	C
3	B	43	D	83	B	123	B	163	B
4	C	44	A	84	B	124	B	164	C
5	D	45	A	85	A	125	C	165	A
6	B	46	B	86	A	126	B	166	B
7	B	47	B	87	A	127	A	167	B
8	A	48	C	88	B	128	A	168	C
9	B	49	C	89	B	129	B	169	C
10	D	50	A	90	A	130	A	170	A
11	A	51	A	91	C	131	B	171	B
12	B	52	B	92	B	132	B	172	C
13	C	53	C	93	A	133	A	173	C
14	A	54	B	94	B	134	A	174	B
15	A, D	55	A	95	A	135	B	175	D
16	D	56	A	96	A	136	B	176	A
17	B	57	A	97	B	137	B	177	B
18	B	58	A	98	A	138	D	178	B
19	B	59	A	99	B	139	A	179	D
20	B	60	A	100	A	140	B	180	C



BELIEVE IN YOURSELF!

Hard work today, bright future tomorrow.
Stay focused, stay consistent and never give up.
Your success is our mission!



JOIN OUR WHATSAPP

Get Daily MCQs, Notes, Past Papers, Updates

↓ CLICK THE LINK BELOW TO JOIN

MDCAT

MOCK TEST

PMDC PATTERN • FULL LENGTH TEST • INSTANT RESULT

PMDC PATTERN

3 HOURS FULL TEST

INSTANT RESULT

DETAILED EXPLANATIONS

EXAM LIKE EXPERIENCE

SUBJECTS
 BIOLOGY • CHEMISTRY
 PHYSICS • ENGLISH

100% REAL MDCAT EXPERIENCE

PRACTICE TODAY SCORE TOMORROW

PAK-MED

STUDY CORNER

YOUR SUCCESS, OUR MISSION

BEST PRACTICE

HIGHER SCORE

BETTER PREPARATION

BRIGHTER FUTURE